Time-Ordered Boundary Monodromy at a Quadratic Band-Merging Map:

Exact Geometric Cycle Determinants and a Nonuniform-Conditioning Barrier

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Abstract

At the first algebraic band-merging parameter of $f_u(x) = 1 - ux^2$, the parity-extracted deterministic bulk determinant has the exact edge factor $(1-z^2/\lambda)^{-1}$. Its canonical degree-N finite section is $\Pi_N(z^2/\lambda) = 1 + z^2/\lambda + \cdots + (z^2/\lambda)^N$, whose reciprocal zeros form the roots-of-unity cloud observed for the Gaussian Markov operator. The preceding endpoint-rank theorem supplied the correct half-logarithmic number of directions but did not supply their time ordering. We construct the missing deterministic time-ordered model and identify a new obstruction to transferring it to the noisy operator.

Let $S = f^2$, and let p_{2k} be the primitive boundary cycle coded by $CA(CB)^{k-1}$. We first show that its own S-orbit is an exact length-k closed chain. This chain, rather than the collection of endpoint points from different periods, is the dynamically invariant object. We then prove the new critical-return multiplier law

$$(S^k)'(p_{2k}) = -C_M\lambda^k(1 + o(1)), \qquad C_M > 0.$$

The exponent is only k, although $S'(r) = \lambda^2$: the final return passes within $O(\lambda^{-k})$ of the quadratic critical point and cancels one power of the repelling expansion.

Weighting the ordered chain by $\omega_{k,j} = |S'(x_{k,j})|^{-1}$ gives a finite cyclic transfer matrix W_k. Put $\rho_k = |(S^k)'(p_{2k})|^{-1/k}$. Every choice of its individual weights is removed by an exact diagonal similarity, and after the canonical bipartite one-step lift one obtains

$$\boxed{\frac{\det(\mathrm{I} - z\mathbf{W}_k)}{1 - \rho_k z^2} = \Pi_{k-1}(\rho_k z^2).}$$

Consequently the edge-deflated reciprocal cloud has exact phases $\pm j\pi/k$ and radius

$$\sqrt{\rho_k} = \lambda^{-1/2}\exp\left[-\frac{\log C_M}{2k} + o(k^{-1})\right].$$

This corrects an earlier "finite unilateral shift" description: a bare finite unilateral shift is nilpotent and has determinant one; the geometric polynomial requires a cyclic return followed by edge deflation.

The correction is not cosmetic. The canonical diagonal balance has condition number at least $c\lambda^k$, and the Euclidean eigenvalue condition numbers are at least $c\lambda^k/k$. At the endpoint-rank scale $k = \log(1/\sigma)/(2\log\lambda) + O(1)$ these bounds become $\Omega(\sigma^{-1/2})$ and $\Omega(\sigma^{-1/2}/\log(1/\sigma))$. Thus a uniformly conditioned Euclidean Feshbach identification is unavailable for the unrenormalized orbit basis; an adapted norm or structured Grushin normalization is needed.

High-precision computations give $C_M = 1.946342905200967\ldots$. With the integer selected independently from the preceding endpoint-row calculation, the resulting finite-k radii are closer than the limiting radius to every archived cloud mean. These spectral comparisons are ordinary floating-point diagnostics. We do not claim that W_k is already a Feshbach block of the noisy Markov operator.

Keywords: transfer operator; weighted cyclic shift; periodic monodromy; Gaussian perturbation; resonance cloud; Grushin problem; non-normal spectrum; quadratic map.

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1 Introduction

The small-noise spectral problem at a band-merging map has now separated into three scales. The peripheral negative resonance is governed by a coupled critical-value boundary layer and satisfies a square-root splitting law [8]. After that mode and the Perron mode are removed, the deterministic bulk determinant has the exact factorization

$$\widehat{D}_{0,\mathrm{bulk},2}(z) = \frac{\mathcal{G}(z)}{1 - z^2/\lambda}, \qquad \mathcal{G}(z) \neq 0 \quad (|z| < \lambda), \tag{1}$$

so locally uniform convergence of the entire noisy determinants through $z = \pm\sqrt{\lambda}$ is impossible [9]. Finally, the endpoint Gaussian row family has an intrinsic threshold rank

$$N_\sigma^{\mathrm{row}} = \frac{\log(1/\sigma)}{2\log\lambda} + O(1), \tag{2}$$

both for Hellinger fingerprints and for normalized linear kernel rows [10].

The missing arrow is dynamical. Singular values count stable row directions, but they do not specify how the transfer operator moves those directions. The archived noisy spectra suggest the geometric section

$$\Pi_N(q) = 1 + q + \cdots + q^N, \qquad q = z^2/\lambda, \tag{3}$$

because its reciprocal zeros have radius $\lambda^{-1/2}$ and phases $\pm j\pi/(N+1)$. The previous route map described this as the characteristic polynomial of a finite unilateral shift. Literally, that statement is false: if U_N is the ordinary finite unilateral shift, then $U_N^N = 0$ and

$$\det(\mathrm{I} - qU_N) = 1. \tag{4}$$

The polynomial Π_N is instead the determinant of a cyclic shift after one distinguished eigenvalue has been removed:

$$\frac{\det(\mathrm{I} - qC_{N+1})}{1 - q} = \frac{1 - q^{N+1}}{1 - q} = \Pi_N(q). \tag{5}$$

This paper replaces the inaccurate unilateral wording by an exact dynamical cyclic return.

The correction does not alter the pole theorem (1), the algebraic finite section (3), or the archived noisy cloud. It changes the operator mechanism that must be proved. A one-way chain is insufficient; one needs a closed time-ordered chain, a distinguished edge channel to deflate, and quantitative control of the basis that balances its weights.

Main results and logical status

Let p_{2k} denote the boundary point constructed from the inverse word $CA(CB)^{k-1}$, and let $S = f^2$. The paper establishes the following.

M1. Exact time order. The S-orbit of p_{2k} is an explicit cyclic chain of length k. In contrast, the endpoint dictionary $\{p_{2k} : k \geq 1\}$ is not forward invariant: every p_{2k} with $k \geq 2$ moves a fixed positive distance away from the entire dictionary in one component step.

M2. Boundary multiplier theorem. If $M_k = (S^k)'(p_{2k})$, then

$$M_k = -C_M \lambda^k (1 + o(1)), \qquad C_M > 0. \tag{6}$$

This is an analytic theorem. The displayed decimal for C_M later in the paper is a high-precision computation, not an interval enclosure.

M3. Exact geometric determinant. The inverse-Jacobian weighted cyclic chain has an edge-deflated bipartite determinant exactly equal to

$$\Pi_{k-1}(\rho_k z^2), \qquad \rho_k = |M_k|^{-1/k}. \tag{7}$$

Thus both the phase grid and the limiting radius follow from the same deterministic time ordering.

M4. Conditioning barrier. The diagonal similarity that balances the cycle has $\mathrm{cond}_2(D_k) \geq c\lambda^k$. The ordinary Euclidean eigenvalue condition number has the lower bound $c\lambda^k/k$. These are analytic statements; they rule out one particular uniformly conditioned reduction strategy, not every adapted-space construction.

M5. Numerical comparison. At the seven archived noises, finite-cycle radii selected from the independent Hellinger or linear row ranks are closer to the cloud means than the limiting value $\lambda^{-1/2}$. This is reproducible floating-point evidence only. No equality between row rank, monodromy dimension, and Markov cloud degree is promoted to a theorem.

The resulting logical chain is

$$\boxed{\begin{array}{c} \text{boundary word} \Longrightarrow \text{exact weighted cyclic monodromy} \Longrightarrow \Pi_{k-1}(\rho_k z^2), \\ \text{noisy Markov Feshbach block} \quad \Longrightarrow ? \quad \text{that monodromy in an adapted norm.} \end{array}} \tag{8}$$

The upper line is proved here. The lower arrow remains the next operator problem.

2 The boundary word and component dynamics

Let u_c be the root in $(1, 2)$ of

$$u^3 - 2u^2 + 2u - 2 = 0, \tag{9}$$

and put

$$f(x) = 1 - u_c x^2, \qquad r = u_c - 1, \qquad \lambda = 2u_c r. \tag{10}$$

Then

$$0 \longmapsto 1 \longmapsto -r \longmapsto r \longmapsto r \tag{11}$$

and

$$u_c = 1.543689012692076\ldots, \quad r = 0.543689012692076\ldots, \quad \lambda = 1.678573510428322\ldots. \tag{12}$$

Write $S = f^2$. The fixed point r has

$$S'(r) = \lambda^2. \tag{13}$$

For $y \in [-r, 1]$, define the two inverse branches

$$g_+(y) = \sqrt{\frac{1-y}{u_c}}, \qquad g_-(y) = -\sqrt{\frac{1-y}{u_c}}, \tag{14}$$

and the two inverse component branches

$$h = g_+ \circ g_+, \qquad q = g_+ \circ g_-. \tag{15}$$

On their corresponding branches,

$$S \circ h = \mathrm{I}, \qquad S \circ q = \mathrm{I}. \tag{16}$$

At r,

$$h(r) = r, \quad h'(r) = a := \lambda^{-2}, \quad q(r) = 1, \quad q'(r) = -\frac{1}{2u_\mathrm{c}\lambda}. \tag{17}$$

For $k \geq 1$, the word $CA(CB)^{k-1}$ has inverse composition

$$G_k = q \circ h^{k-1}. \tag{18}$$

Its unique fixed point is denoted by

$$p_k := p_{2k} = G_k(p_k), \qquad \delta_k = 1 - p_k. \tag{19}$$

The subscript k will always mean component time; the physical period is $2k$. The boundary-crowding theorem gives

$$\delta_k = C_\mathrm{b} a^k (1 + o(1)) = C_\mathrm{b} \lambda^{-2k}(1 + o(1)), \qquad C_\mathrm{b} > 0 \tag{20}$$

[10, 11].

The proof below uses the Koenigs coordinate ψ of h at r, normalized by

$$\psi(h(x)) = a\psi(x), \qquad \psi(r) = 0, \qquad \psi'(r) = 1. \tag{21}$$

The coordinate is analytic near r and extends along the real attracting basin. Although h' is singular at 1, the point

$$b := h(1) = u_\mathrm{c}^{-1/2} \tag{22}$$

is interior, so $0 < \psi'(b) < \infty$.

3 The exact time-ordered return chain

The endpoint points p_k come from different periodic orbits. Time evolution does not move from p_k to p_{k-1}. The correct chain lies inside one boundary orbit.

Theorem 3.1 (Exact boundary-word time order)**.** *For $k \geq 1$, define*

$$x_{k,0} = p_k, \qquad x_{k,j} = h^{k-j}(p_k), \quad 1 \leq j \leq k-1. \tag{23}$$

Then these are the k distinct points of the S-cycle of p_k, in forward time order:

$$S(x_{k,j}) = x_{k,j+1} \quad (0 \leq j < k-1), \qquad S(x_{k,k-1}) = x_{k,0}. \tag{24}$$

Proof. The fixed-point identity in (19) is $p_k = q(h^{k-1}(p_k))$. Applying $S \circ q = \mathrm{I}$ gives

$$S(p_k) = h^{k-1}(p_k) = x_{k,1}.$$

For $1 \leq j < k-1$, apply $S \circ h = \mathrm{I}$ to obtain

$$S(h^{k-j}(p_k)) = h^{k-j-1}(p_k) = x_{k,j+1}.$$

The same identity at the final point gives $S(h(p_k)) = p_k$. Distinctness follows from primitivity of $CA(CB)^{k-1}$. □

The first internal point lies in the repelling-boundary layer. Its scale is the same as the endpoint clearance.

Corollary 3.2 (Endpoint-to-repelling scale bridge)**.** *As $k \to \infty$,*

$$x_{k,1} - r = \frac{\delta_k}{-q'(r)}(1 + o(1)) = 2u_c\lambda\, \delta_k(1 + o(1)). \tag{25}$$

In particular, the endpoint ladder across different periods and the first internal levels of the ordered cycles have the same λ^{-2k} resolution clock.

Proof. By (23) and (19), $p_k = q(x_{k,1})$. Since $x_{k,1} \to r$ and $q(r) = 1$, Taylor expansion gives

$$\delta_k = 1 - q(x_{k,1}) = -q'(r)(x_{k,1} - r)(1 + o(1)).$$

Insert (17). □

3.1 Why the cross-period endpoint dictionary cannot be the shift

The failure of direct endpoint time ordering is separated from the later conditioning issue.

Proposition 3.3 (Fixed gap after one component step)**.** *The points p_k increase with k. For every $k \geq 2$,*

$$S(p_k) < b = h(1) < p_1 \leq p_j \qquad (j \geq 1). \tag{26}$$

Thus $S(p_k)$ is separated from the entire cross-period endpoint dictionary by the fixed gap

$$d_* = p_1 - b = 0.096158898694365\ldots > 0. \tag{27}$$

Proof. On $(r, 1)$, h is increasing and $h(x) < x$, while q is decreasing. Hence $G_{k+1}(x) = q(h^k(x)) > q(h^{k-1}(x)) = G_k(x)$. Each G_k is decreasing, so the unique zero of $G_k(x) - x$ moves to the right: $p_{k+1} > p_k$. Moreover, $p_1 < 1$ and the explicit formula for q gives

$$p_1 = q(p_1) = \sqrt{\frac{1 + \sqrt{(1 - p_1)/u_c}}{u_c}} > u_c^{-1/2} = b.$$

For $k \geq 2$, theorem 3.1 and monotonicity of h give

$$S(p_k) = h^{k-1}(p_k) \leq h(p_k) < h(1) = b.$$

The monotonicity of p_j gives the remaining inequalities. The decimal is a high-precision evaluation of $p_1 - b$. □

Remark 3.4 (Consequence for a localized compression)*.* A family of packets concentrated at the points p_k cannot be permuted by one application of the two-step dynamics: all nontrivial packets move into a neighborhood of $[r, b]$, a fixed distance from their original centers. Consequently the RH-16 endpoint row dictionary can supply a dimension count, but it cannot by itself be the time-ordered basis. Any noisy Feshbach block must include the internal chain (23) and the coupling between the endpoint and repelling-boundary layers.

4 A critical-return multiplier theorem

Let

$$M_k = (S^k)'(p_k) = \prod_{j=0}^{k-1} S'(x_{k,j}). \tag{28}$$

If every factor were asymptotic to $S'(r) = \lambda^2$, one might expect $|M_k|$ to grow like λ^{2k}. The final factor is instead exponentially small because $x_{k,k-1} = h(p_k)$ maps through a point close to the critical point.

Lemma 4.1 (Endpoint derivative of the inverse branch)**.** *As $\delta \downarrow 0$,*

$$h'(1-\delta) = \frac{1}{4u_{\mathrm{c}}\sqrt{\delta}}(1 + O(\sqrt{\delta})). \tag{29}$$

Consequently

$$h'(p_k) = \frac{\lambda^k}{4u_{\mathrm{c}}\sqrt{C_{\mathrm{b}}}}(1 + o(1)). \tag{30}$$

Proof. Differentiating $h = g_+ \circ g_+$ gives

$$h'(x) = \frac{1}{4u_{\mathrm{c}}^2 h(x) g_+(x)}. \tag{31}$$

For $x = 1-\delta$, $g_+(x) = \sqrt{\delta/u_{\mathrm{c}}}$ and $h(x) = u_{\mathrm{c}}^{-1/2} + O(\sqrt{\delta})$. Substitution proves (29). Apply (20) and $a^{-k/2} = \lambda^k$ to obtain (30). ☐

Theorem 4.2 (Critical-return boundary multiplier law)**.** *There is a constant $C_M > 0$ such that*

$$\boxed{M_k = -C_M \lambda^k (1 + o(1)).} \tag{32}$$

With $b = u_{\mathrm{c}}^{-1/2}$ and the normalization (21), the constant is

$$C_M = \frac{4u_{\mathrm{c}}\sqrt{C_{\mathrm{b}}}}{|q'(r)|\psi'(b)\lambda^4}. \tag{33}$$

Proof. The map $G_k = q \circ h^{k-1}$ is the inverse branch of S^k that fixes p_k. Therefore

$$M_k = \frac{1}{G_k'(p_k)}. \tag{34}$$

Separate the singular first derivative of h^{k-1}:

$$(h^{k-1})'(p_k) = h'(p_k)(h^{k-2})'(h(p_k)). \tag{35}$$

Differentiating the Koenigs identity $\psi(h^n(x)) = a^n \psi(x)$ gives

$$(h^n)'(x) = a^n \frac{\psi'(x)}{\psi'(h^n(x))}. \tag{36}$$

Use this with $n = k-2$ and $x = h(p_k)$. Since $h(p_k) \to b$, $h^{k-1}(p_k) \to r$, and $\psi'(r) = 1$,

$$(h^{k-2})'(h(p_k)) = a^{k-2}\psi'(b)(1 + o(1)). \tag{37}$$

Combining lemma 4.1 with (37) gives

$$\begin{aligned}(h^{k-1})'(p_k) &= \frac{\psi'(b)}{4u_\mathrm{c}\sqrt{C_\mathrm{b}}} a^{k-2-k/2}(1+o(1)) \\ &= \frac{\psi'(b)\lambda^4}{4u_\mathrm{c}\sqrt{C_\mathrm{b}}}\lambda^{-k}(1+o(1)). \end{aligned} \tag{38}$$

Moreover, $q'(h^{k-1}(p_k)) = q'(r)(1+o(1))$. Hence

$$G_k'(p_k) = q'(r)\frac{\psi'(b)\lambda^4}{4u_\mathrm{c}\sqrt{C_\mathrm{b}}}\lambda^{-k}(1+o(1)).$$

The derivative is negative. Inverting it proves (32) and (33). □

Remark 4.3 (Where the missing power goes). The final point of the ordered chain is $x_{k,k-1} = h(p_k)$. From $S \circ h = \mathrm{I}$,

$$S'(h(p_k)) = \frac{1}{h'(p_k)} = 4u_\mathrm{c}\sqrt{C_\mathrm{b}}\,\lambda^{-k}(1+o(1)). \tag{39}$$

The other $k-1$ factors collectively carry the repelling expansion. The single critical return (39) removes one exponential power, leaving $|M_k| \asymp \lambda^k$ rather than λ^{2k}.

5 Weighted cyclic monodromy and the geometric determinant

The physical flat trace is associated with the inverse-Jacobian stability channel [1]. Along the ordered cycle, define

$$\omega_{k,j} = \frac{1}{|S'(x_{k,j})|}, \qquad 0 \le j \le k-1, \tag{40}$$

and let W_k act on the standard basis of $\mathbb{C}^k$ by

$$W_k e_j = \omega_{k,j} e_{j+1 \pmod k}. \tag{41}$$

Its total weight is

$$\prod_{j=0}^{k-1} \omega_{k,j} = |M_k|^{-1}. \tag{42}$$

Put

$$\rho_k = |M_k|^{-1/k}. \tag{43}$$

Theorem 5.1 (Exact balancing and cyclic determinant)**.** *There is a positive diagonal matrix D_k such that*

$$D_k^{-1} W_k D_k = \rho_k C_k, \tag{44}$$

where $C_k e_j = e_{j+1 \pmod k}$ is the unweighted cyclic shift. Consequently

$$W_k^k = \rho_k^k \mathrm{I}, \tag{45}$$

$$\mathrm{spec}(W_k) = \{\rho_k e^{2\pi i \ell/k} : 0 \le \ell < k\}, \tag{46}$$

$$\det(\mathrm{I} - wW_k) = 1 - (\rho_k w)^k. \tag{47}$$

Proof. Choose $d_{k,0} > 0$ and define cyclically

$$d_{k,j+1} = \frac{\omega_{k,j}}{\rho_k} d_{k,j}. \tag{48}$$

The closure condition at $j = k-1$ holds because $\prod_j \omega_{k,j} = \rho_k^k$. With $D_k = \mathrm{diag}(d_{k,0}, \dots, d_{k,k-1})$, each balanced matrix entry equals ρ_k, proving (44). The remaining claims are the elementary spectrum and determinant of C_k. □

The original map exchanges two components in one step. The canonical bipartite lift of the component matrix is

$$\mathbf{W}_k = \begin{pmatrix} 0 & \mathrm{I} \\ W_k & 0 \end{pmatrix} \quad \text{on } \mathbb{C}^k \oplus \mathbb{C}^k. \tag{49}$$

Then $\mathbf{W}_k^2 = \operatorname{diag}(W_k, W_k)$ and a Schur complement gives

$$\det(\mathrm{I} - z\mathbf{W}_k) = \det(\mathrm{I} - z^2 W_k) = 1 - (\rho_k z^2)^k. \tag{50}$$

Remark 5.2 (Algebraic normalization of the lift). The choice of an identity in the upper-right block is a normalization, not a claim about the individual one-step branch weights. More generally, if $A_k B_k$ and $B_k A_k$ are the two component return matrices, then

$$\det\begin{pmatrix} \mathrm{I} & -zA_k \\ -zB_k & \mathrm{I} \end{pmatrix} = \det(\mathrm{I} - z^2 B_k A_k).$$

Thus the determinant depends only on the two-step monodromy. Identifying the separate blocks with the noisy one-step dynamics remains part of the Feshbach problem in section 7.

Corollary 5.3 (Exact edge-deflated geometric section)**.** *After removing the distinguished real edge pair* $\pm\sqrt{\rho_k}$*,*

$$\boxed{\frac{\det(\mathrm{I} - z\mathbf{W}_k)}{1 - \rho_k z^2} = \Pi_{k-1}(\rho_k z^2).} \tag{51}$$

The reciprocal resonances of the quotient are exactly

$$\mu_{k,j}^{\pm} = \sqrt{\rho_k}\exp\left(\pm\frac{ij\pi}{k}\right), \qquad 1 \le j \le k-1. \tag{52}$$

Proof. Divide (50) by $1 - \rho_k z^2$ and use the finite geometric identity. The zeros of $\Pi_{k-1}(q)$ are the nontrivial kth roots of unity. Taking both square roots and then reciprocals gives (52). □

The radius now follows from the multiplier theorem.

Corollary 5.4 (Finite-radius law)**.** *As* $k \to \infty$*,*

$$\rho_k = \lambda^{-1}\exp\left[-\frac{\log C_M}{k} + o(k^{-1})\right], \tag{53}$$

$$r_k := \sqrt{\rho_k} = \lambda^{-1/2}\exp\left[-\frac{\log C_M}{2k} + o(k^{-1})\right]. \tag{54}$$

Thus (51) *tends coefficientwise to the RH-15 section* $\Pi_{k-1}(z^2/\lambda)$*.*

Proof. Take logarithms in $\rho_k = |M_k|^{-1/k}$ and use theorem 4.2. The second identity is its square root. □

5.1 Cyclic return versus unilateral truncation

The distinction can be summarized algebraically. Let U_k be the nilpotent shift $U_k e_j = e_{j+1}$ for $j < k-1$ and $U_k e_{k-1} = 0$. Then

$$\det(\mathrm{I} - qU_k) = 1. \tag{55}$$

The resolvent of U_k contains a finite geometric sum,

$$(\mathrm{I} - qU_k)^{-1} = \mathrm{I} + qU_k + \cdots + q^{k-1}U_k^{k-1}, \tag{56}$$

but that sum is a matrix element or transfer function, not its determinant. Closing the final arrow turns U_k into C_k; only then does edge deflation produce Π_{k-1}. The boundary word supplies precisely this closing arrow through $S(x_{k,k-1}) = x_{k,0}$.

Remark 5.5 (What the edge pair means). The factor $1-\rho_k z^2$ in (51) is the distinguished channel of the cycle model. It is not the Perron/parity pair ± 1 of the deterministic Markov dynamics. Rather, its limit is the formal edge pair $\pm\lambda^{-1/2}$ associated with the pole $(1-z^2/\lambda)^{-1}$. The geometric cloud is the determinant left after that edge channel is removed.

6 The nonuniform-conditioning barrier

The spectrum in (46) depends only on the product of the weights. Perturbative stability depends on their distribution. Here the critical return puts an exponentially large inverse-Jacobian weight on one edge of the cycle.

Normalize D_k by an arbitrary positive scalar and let

$$\kappa_k = \operatorname{cond}_2(D_k) = \frac{\max_j d_{k,j}}{\min_j d_{k,j}}. \tag{57}$$

The last recurrence in (48) gives

$$\frac{d_{k,0}}{d_{k,k-1}} = \frac{\omega_{k,k-1}}{\rho_k}. \tag{58}$$

Theorem 6.1 (Exponential balancing barrier)**.** *There are $c>0$ and k_0 such that*

$$\boxed{\kappa_k \ge c\lambda^k \qquad (k\ge k_0).} \tag{59}$$

More precisely,

$$\omega_{k,k-1} = \frac{\lambda^k}{4u_{\mathrm{c}}\sqrt{C_{\mathrm{b}}}}(1+o(1)), \tag{60}$$

and therefore

$$\frac{d_{k,0}}{d_{k,k-1}} = \frac{\lambda^{k+1}}{4u_{\mathrm{c}}\sqrt{C_{\mathrm{b}}}}(1+o(1)). \tag{61}$$

Proof. The final ordered point is $x_{k,k-1}=h(p_k)$. Since $S\circ h=\mathrm{I}$,

$$\omega_{k,k-1} = \frac{1}{|S'(h(p_k))|} = h'(p_k).$$

Equation (60) is therefore (30). Combine it with $\rho_k\to\lambda^{-1}$ in (58). Since the condition number dominates the ratio of any two diagonal entries, (59) follows. □

The same imbalance appears in the ordinary eigenvalue condition numbers. Let f_ℓ be a unit Fourier eigenvector of C_k. From (44), right and left eigenvectors of W_k may be chosen as $D_k f_\ell$ and $D_k^{-1} f_\ell$. Their common Euclidean eigenvalue condition number is

$$\chi_k = \frac{\|d_k\|_2\|d_k^{-1}\|_2}{k}, \tag{62}$$

where $d_k=(d_{k,0},\dots,d_{k,k-1})$ and the inverse is componentwise.

Corollary 6.2 (Eigenvalue sensitivity at the noise clock)**.** *For all sufficiently large k,*

$$\chi_k \ge \frac{\kappa_k}{k} \ge \frac{c\lambda^k}{k}. \tag{63}$$

If

$$k_\sigma = \frac{\log(1/\sigma)}{2\log\lambda} + O(1), \tag{64}$$

then

$$\kappa_{k_\sigma} = \Omega(\sigma^{-1/2}), \tag{65}$$

$$\chi_{k_\sigma} = \Omega\left(\frac{\sigma^{-1/2}}{\log(1/\sigma)}\right). \tag{66}$$

Proof. The two sums in (62) contain respectively $\max_j d_{k,j}$ and $(\min_j d_{k,j})^{-1}$, proving the first inequality. Equation (64) gives $\lambda^{k_\sigma} = \Theta(\sigma^{-1/2})$ and $k_\sigma = \Theta(\log(1/\sigma))$. □

Remark 6.3 (Interpretation, not an impossibility theorem). The usual Bauer–Fike estimate would multiply a Euclidean operator-norm remainder by an exponentially growing eigenvector condition number [4, 7]. Therefore a remainder merely small on the k^{-1} phase-spacing scale is not enough for this unbalanced basis. This does not prove that the noisy cloud is unstable. The perturbation is structured, the Gaussian row Gram matrix is nontrivial, and an adapted anisotropic norm can absorb part of the diagonal balance. The theorem says that uniform Euclidean pre- and post-conditioners cannot be assumed for the raw orbit packets.

7 Relation to the noisy Gaussian operator

For $\sigma > 0$, let $\mathcal{K}_\sigma$ be the row-normalized folded Gaussian Markov operator on $[0, 1]$ used in the preceding papers. It is compact and strongly positive, with Perron eigenvalue one and a simple negative parity resonance. After those two modes are removed, its bulk spectrum contains the cloud compared below.

The matrix W_k is not defined by projecting $\mathcal{K}_\sigma^2$ onto computed eigenvectors. It is a deterministic first-stability-channel monodromy built from an explicit periodic word. This independence makes the phase theorem noncircular, but it also leaves a genuine proof obligation.

7.1 What has now been proved

The following parts of the proposed shift mechanism are exact.

(i) The half-logarithmic row count has the same geometric scale as the first internal point of the time-ordered boundary cycle, by corollary 3.2.

(ii) The relevant time ordering is cyclic and is supplied by one primitive boundary orbit, by theorem 3.1.

(iii) The inverse-Jacobian stability channel of that orbit gives the exact polynomial $\Pi_{k-1}(\rho_k z^2)$, by corollary 5.3.

(iv) The limiting radius and its first $1/k$ correction follow from a new multiplier theorem, not from fitting the noisy cloud.

7.2 What has not been proved

No theorem in this paper asserts

$$\text{Feshbach block of } \mathcal{K}_\sigma \simeq \mathbf{W}_{k_\sigma}. \tag{67}$$

In particular, we do not prove equality among

$$N_\sigma^{\text{Hellinger}}, \qquad N_\sigma^{\text{linear}}, \qquad N_\sigma^{\text{cloud}}. \tag{68}$$

All three obey, or numerically follow, the same leading clock, but their bounded staircase phases need not agree at finite noise.

The cross-period endpoint basis cannot fix this problem: by proposition 3.3, its direct two-step compression misses the ordered chain. Conversely, the raw orbit basis has the exponential conditioning in theorem 6.1. These two facts narrow the next construction considerably.

7.3 A corrected Grushin route

A viable operator proof should contain four estimates.

G1. Paired boundary extraction. Remove the RH-14 parity profile at r and the critical-value endpoint layer at 1 before forming the bulk Schur complement [8].

G2. Ordered transition space. Build packets along the internal points $x_{k,j}$ of one boundary word, rather than using only the means p_k from different words. Use corollary 3.2 to compare the resulting approximation numbers with the RH-16 endpoint row operator.

G3. Adapted balance. Incorporate D_k into the norm or into the entrance/exit maps of a Grushin problem. Uniform conditioning in the unweighted Euclidean packet norm is contradicted by (65); the correct statement must be weighted or exploit the structured Gaussian remainder [6].

G4. Residual normality. After extracting the resulting cloud determinant, prove local bounds for the residual determinant on compact subsets of $|z| < \lambda$. This remains necessary to identify the nonzero factor $\mathcal{G}$ in (1).

This route is more constrained than the previous "unilateral shift plus a small remainder" formulation. It specifies the closure edge, the edge factor to deflate, the natural finite-radius correction, and the norm problem that any perturbation argument must solve.

8 High-precision and archived-spectrum audit

The analytic results above require no numerical hypothesis. The computations serve three narrower purposes: evaluate the constants, verify the exact finite identities in an independent implementation, and compare the parameter-free finite-k radii with already archived noisy spectra.

8.1 Method and reproducibility

The fixed point of $G_k = q \circ h^{k-1}$ is evaluated by contraction with 130 decimal digits. The multiplier is computed in two independent ways:

$$M_k = \prod_{j=0}^{k-1} S'(x_{k,j}) = \frac{1}{G_k'(p_k)}. \tag{69}$$

The test suite checks the equality, the exact cyclic time order, the diagonal similarity, the bipartite determinant, the reciprocal roots, and the conditioning bounds. NumPy, mpmath, and Matplotlib provide the implementation [2, 3, 5].

The calculation gives

$$C_M = 1.946342905200967\ldots. \tag{70}$$

The value of $|M_k|/\lambda^k$ is stable to the displayed digits by $k = 80$. This is a multiprecision diagnostic, not a directed-rounding certificate.

Table 1: Boundary-cycle multiplier, finite two-step radius, one-step radius, and balancing. Here $\rho_k = |M_k|^{-1/k}$ and $r_k = \sqrt{\rho_k}$.

k	δ_k	$\lvert M_k\rvert/\lambda^k$	ρ_k	r_k	$\mathrm{cond}_2(D_k)/\lambda^k$
4	6.495×10^{-3}	1.8425567	0.5113337	0.7150760	0.511913
8	1.142×10^{-4}	1.9372562	0.5484806	0.7405947	0.440383
20	4.633×10^{-10}	1.9463261	0.5762340	0.7591008	0.414032
40	4.659×10^{-19}	1.9463429	0.5859076	0.7654460	0.407186
80	4.710×10^{-37}	1.9463429	0.5908053	0.7686386	0.403800
100	4.736×10^{-46}	1.9463429	0.5917898	0.7692787	0.403138

Figure 1 displays the two asymptotic mechanisms. The scaled multiplier rapidly converges, while the balancing and eigenvalue condition numbers grow exponentially. The lower-right panel shows the exact $k = 8$ time order: the endpoint first jumps into an exponentially thin layer above r, then moves outward along the inverse-branch ladder before the critical return closes the cycle.

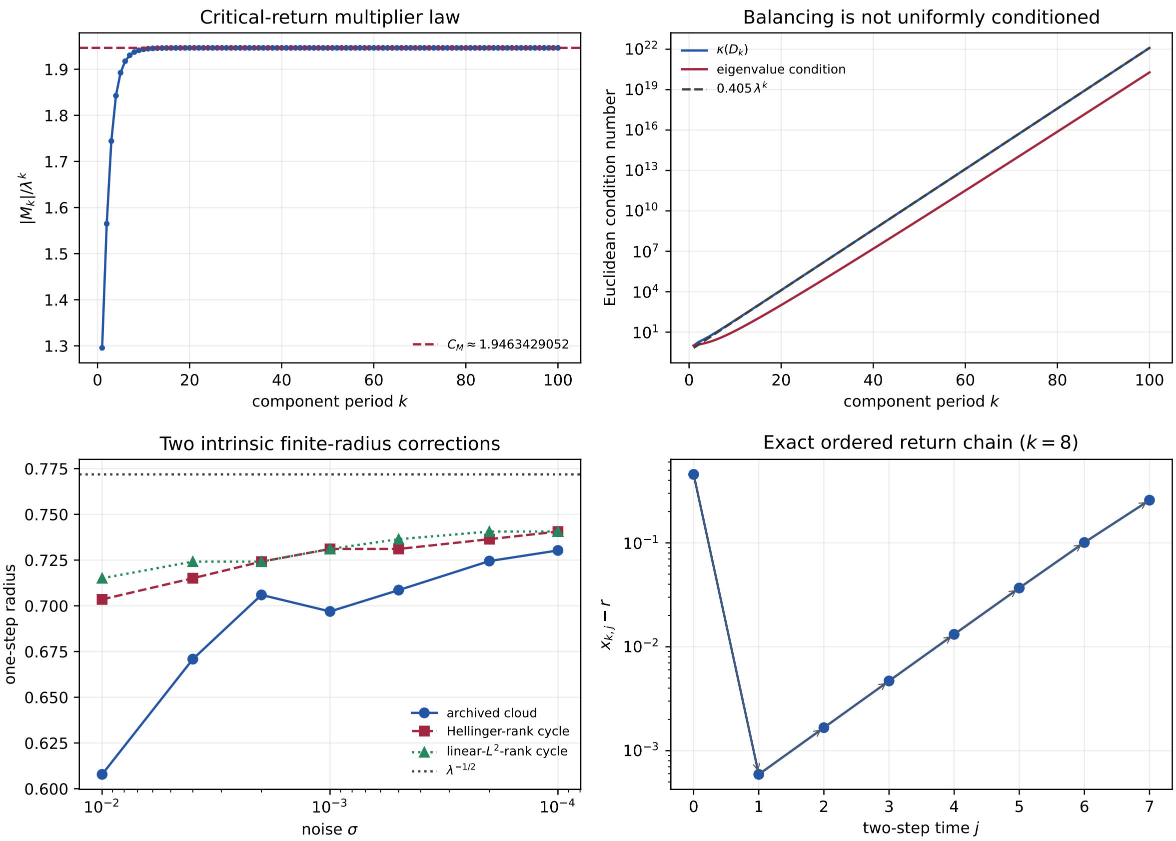

Figure 1: Time-ordered boundary monodromy audit. Top left: the multiplier law $|M_k| \sim C_M \lambda^k$. Top right: the canonical diagonal balance and the ordinary eigenvalue condition number; the dashed line is a numerical λ^k reference. Bottom left: archived noisy cloud radii and the two finite-cycle predictions selected independently from the Hellinger and linear-row ranks. Bottom right: the exact two-step order of the $k = 8$ boundary cycle, shown by distance from the repelling boundary r.

8.2 Two intrinsic integer choices

At each archived noise, we recompute the RH-16 half-energy counts $s_j^2 > 1/2$ without reading any Markov eigenvalue. The Hellinger and linear row families have the same leading rank theorem but different bounded staircase phases. For each integer N, the cycle model uses $k = N + 1$ and predicts $r_k = |M_k|^{-1/(2k)}$.

Table 2: Intrinsic row degrees, archived cloud degree and mean radius, and the two finite-cycle radius predictions. The phase RMS is measured against the common roots-of-unity grid.

σ	N_H	N_L	N_{cloud}	$\bar{r}_{\mathrm{cloud}}$	r_{N_H+1}	r_{N_L+1}	phase RMS
10^{-2}	2	3	3	0.607841	0.703501	0.715076	0.07933
4×10^{-3}	3	4	3	0.670849	0.715076	0.724152	0.07773
2×10^{-3}	4	4	4	0.705958	0.724152	0.724152	0.02431
10^{-3}	5	5	5	0.696908	0.731090	0.731090	0.02184
5×10^{-4}	5	6	5	0.708586	0.731090	0.736423	0.04522
2×10^{-4}	6	7	6	0.724468	0.736423	0.740595	0.02636
10^{-4}	7	7	7	0.730269	0.740595	0.740595	0.01124

The Hellinger degree equals the archived cloud degree at six of seven points; the linear-row degree does so at four of seven. Every discrepancy is one. This is consistent with, but does not sharpen, the analytic $O(1)$ rank ambiguity. It also prevents a claim that the transfer-relevant linear rank has already identified the cloud count.

The finite-cycle correction is nevertheless robust. The root-mean-square radial errors over the seven points are

$$\begin{array}{c|l} \text{radius model} & \text{RMS error} \\ \hline \lambda^{-1/2} & 0.0886635 \\ r_{N_H+1} & 0.0436908 \\ r_{N_L+1} & 0.0492538 \\ r_{N_{\mathrm{cloud}}+1} & 0.0473750 \end{array} \tag{71}$$

The last line is conditional on the spectrally selected degree and is listed only as a diagnostic. The first two finite-cycle lines use no cloud amplitude or phase fit. All three finite choices improve the limiting radius at every point in the table.

At $\sigma = 10^{-4}$, both intrinsic ranks and the cloud selection give $N = 7$. The deterministic $k = 8$ cycle predicts

$$r_8 = 0.7405947445\ldots, \qquad \bar{r}_{\mathrm{cloud}} = 0.7302687845\ldots, \qquad \lambda^{-1/2} = 0.7718445063\ldots. \tag{72}$$

The finite prediction has absolute error 0.01033, compared with 0.04158 for the limiting radius.

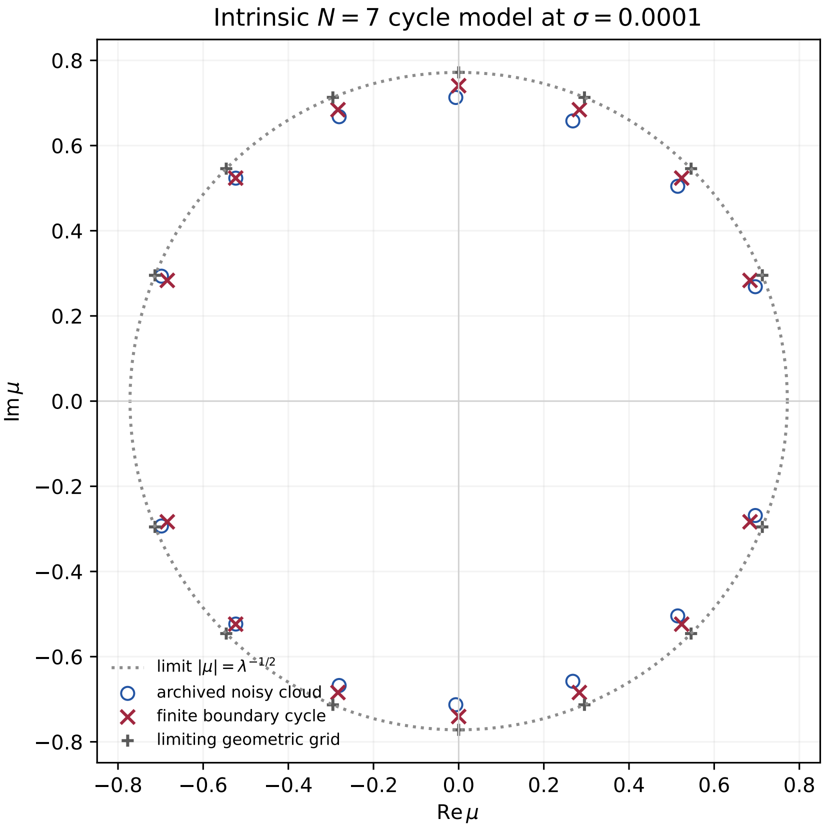

Figure 2: The smallest-noise cloud after the integer $N = 7$ is selected independently by both endpoint-row geometries. Open circles are archived Gaussian Markov resonances. Crosses are the exact edge-deflated $k = 8$ boundary-cycle monodromy, with no radial or phase fit. Plus signs show the same geometric phases at the limiting radius $\lambda^{-1/2}$. The phase agreement and finite-radius improvement are floating-point diagnostics, not a Feshbach theorem.

8.3 Numerical status

The boundary fixed points, multipliers and determinant identities are computed with arbitrary-precision arithmetic. The tests compare the product and inverse-branch multipliers to more than 70 decimal places and verify the matrix identities at double precision. None of these calculations uses interval arithmetic, so the decimal constants are not computer-assisted theorems.

The noisy cloud means and phases are imported unchanged from the RH-15 sparse ARPACK computation with up to 204800 states. The endpoint ranks are recomputed from the exact RH-16 Gram kernels. No noisy transfer matrix is rebuilt in the present audit, and no eigenvalue is selected after examining the cycle prediction. Source hashes, software versions, all rows and both figures are archived with the manuscript.

9 Discussion

9.1 What the geometric cloud now explains

The roots-of-unity phases no longer enter solely as the canonical Taylor section of a pole. They also arise from an explicit dynamical time order. The boundary word gives a closed component-time chain, inverse-Jacobian weights give its first stability channel, and edge deflation gives the geometric polynomial exactly. Arbitrary variation among the local weights does not move the phases because every nonzero weighted cyclic shift is diagonally similar to a constant cyclic shift.

The same construction explains the spectral edge. The multiplier theorem shows that the geometric mean inverse Jacobian tends to λ^{-1} in two-step time, and the bipartite lift produces the one-step radius $\lambda^{-1/2}$. The finite correction depends only on C_M and k; its improvement over the limiting radius is therefore a genuine out-of-sample diagnostic once k is chosen from row geometry.

9.2 Why this is still not the noisy-cloud theorem

There are two obstructions, not one. First, the cross-period endpoint row dictionary is not dynamically invariant. It counts scales correctly but does not contain the internal time chain. Second, the exact cyclic chain is strongly non-normal in the raw Euclidean basis because one critical-return weight is exponentially large. A direct orthogonal compression can miss the chain, while a naive biorthogonal compression can magnify its remainder.

The numerical phase grid suggests that Gaussian smoothing and endpoint coalescence supply a structured regularization of this imbalance. Proving that statement requires controlling the RH-16 row Gram matrix together with the RH-14 paired boundary layers. The conditioning theorem says exactly where the adapted norm must enter; it does not license replacing that proof by an eigenvalue plot.

9.3 No implication for zeta zeros

Everything here concerns one explicit quadratic map, its transfer weights, and a finite spectral-edge model. The geometric phases are roots of unity because the time chain is cyclic. They are not asserted to be Riemann zeros, eigenvalues of a self-adjoint Hilbert–Pólya operator, or evidence for the Riemann hypothesis. Any later arithmetic comparison would require a separate trace formula and a separate operator construction.

10 Conclusion

The time-ordering gate has a precise partial solution. The distinguished boundary word $CA(CB)^{k-1}$ contains an exact length-k component cycle. Its multiplier obeys

$$(S^k)'(p_{2k}) = -C_M \lambda^k (1 + o(1)),$$

where the reduced exponent comes from one exponentially weak critical return. The corresponding inverse-Jacobian weighted cycle has an exact edge-deflated bipartite determinant

$$\Pi_{k-1}(\rho_k z^2), \qquad \rho_k = |(S^k)'(p_{2k})|^{-1/k},$$

and therefore reproduces both the roots-of-unity phases and the limiting radius of the parity-extracted cloud.

The mechanism is cyclic, not unilateral. That correction also reveals the next barrier: balancing the critical-return weight costs at least $c\lambda^k$, or $c\sigma^{-1/2}$ at the endpoint-rank clock. The next paper must therefore construct the noisy Grushin block in an adapted norm, include the full internal chain, and prove residual determinant normality. Until that is done, the monodromy is an exact deterministic realization and a successful numerical predictor, not an identified Markov spectral block.

Data and code availability

The manuscript, high-precision inverse-branch construction, weighted-cycle matrices, tests, CSV and JSON audits, figures, archived-spectrum comparison and source hashes are available with this paper [12].

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